

The Palm OS® Emulator

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3Com
More connected.

Overview

PalmSource 99

- What is it?
- What is it used for?
- Past development
- State of the union
- Future directions
- Report card

Emulators Are Not Slippery Nor Do They Fly…

PalmSource 99

- ...i.e., the first syllable is pronounced as in empty, not as in eel or emu.

  — Dad

What Is It?

PalmSource 99

- Hardware emulator
  - CPU (68000 derivative)
  - Memory
  - Hardware registers

What Is It Used for?

- Development
- Testing
- Demos
- Documentation

The Past

- Originally Copilot
- Adopted by Palm Computing in early 1998
- 2.0b3 released in May 1998
- 2.1 “d” releases started late 1998
  - 2.1d19 posted on 10/30/1998
  - 2.1d26 source posted on 2/26/1999
  - 2.1d27 Unix version posted 5/13/1999

The Present

PalmSource 99

- 2.1 changes since last year
  - Better serial port emulation
  - Profiling output on non-Mac platforms
  - HotSync® user name support
  - Palm OS 1.0 support
  - Better .prc, .pdb, and .pqa file validation
  - Form/form object validation
  - NetLib redirection
  - Network HotSync support

The Present (Cont.)

- 2.1 changes since last year
  - Logging facilities
    - Error Messages
    - Warning Messages
    - Posted events
    - Received events
    - System calls
    - Serial Activity
    - Serial Data
    - NetLib Activity
    - NetLib Data

The Present (Cont.)

- 2.1 changes since last year
  - Autoload, Autorun, AutorunAndQuit directories
  - Palm IIIx, V, and VII support
  - Color device support
  - Conditional breakpoints
  - Complete state saved in session file
  - Compressed session files

The Present (Cont.)

PalmSource 99

- 2.1 changes since last year
  - Support for moving session files to different machines (where the ROM file may be in a different location)
  - 4-bit grayscale support on EZ
  - Unix version
  - New case graphics

The Present (Cont.)

PalmSource 99

- 2.1 changes since last year
  - Command-line options
    - psf <filename>
    - rom <filename>
    - ram <size>
    - device <type>
    - silkscreen <type>

The Present (Cont.)

PalmSource 99

- 3.0!
  - Preferences stored as text files
  - Install Start Menu shortcut on Windows
  - Skins
  - Instruction-level profiling
  - More command-line options
    - skin <name>
    - ram_size <size in K>
    - load_apps <file name list>
    - run_app <app name>
    - quit_on_exit
    - log_save_dir <path>
    - horde <num>, et al

The Present (Cont.)

PalmSource 99

- 3.0!
  - ROM Binder
    - (Demo)
  - Gremlin Hordes
    - (Demo)
  - RPC/Scripting
    - (Demo)

The Future

- Component architecture
- Code coverage
- Debugging displays
  - Logging information
  - Heap display

The Future (Cont.)

PalmSource 99

- Net Hordes
- Exchange data with host clipboard
- Exchange Manager support
- Sound support
- RAM Transfer.prc

Last Year's Future Directions

PalmSource 99

- Scripting
- Networking/IR
- Gremlin Hordes
- Even better error detection
- Direct HotSync support
- Future devices

Last Year's Questions

PalmSource 99

- Urge networking support
- Source code for 2.1
- Problems when crashing/restarting
- Show power consumption
- Symbol support

We Listen, So Talk

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